

Specific dates t_{DEM}

Surface & Bed DEMs

Elmer/Ice (with $A_s \tau_b^m = u_s$)

Stress & velocity fields

Empirical relationships

$$\tau_b(t_{\text{DEM}}) = f_1(H(t_{\text{DEM}})) \quad \& \quad u_{\text{def}}(t_{\text{DEM}}) = f_2(H(t_{\text{DEM}}))$$

+

Annual timeseries t_{Stake}

Field observations

thickness $H(t_{\text{Stake}})$ & surface velocity $u_s(t_{\text{Stake}})$

Temporal interpolation of basal shear stress & sliding velocity

$$\tau_b(t_{\text{Stake}}) = \underline{f_1}(\underline{H(t_{\text{Stake}})}) \quad \& \quad u_b(t_{\text{Stake}}) = \underline{u_s(t_{\text{Stake}})} - \underline{f_2}(\underline{H(t_{\text{Stake}})})$$

Friction Law : $\tau_b = f(u_{\text{bed}})$